1 Core-Hole Potential

Instead of an explicit calculating of the overlap term, S_{pq}, these can also be obtained in terms of the core-hole potential acting on the initial wave function,

$$S_{pq} \approx \langle \phi_p | \hat{v}_{\mathrm{ch}} | \phi_q \rangle \tag{1}$$

with ϕ_p and ϕ_q the electron and hole orbitals of the initial state. We wish to compute the matrix-element,

$$\langle \phi_p | \hat{v}_{\mathrm{ch}} | \phi_q \rangle \approx \int dr \;\; \phi_p^*(\mathbf{r}) v_{ch}(\mathbf{r}) \phi_q(\mathbf{r}), \tag{2}$$

with,

$$v_{ch}(\mathbf{r}) = -\int d\mathbf{r}' \;\; \frac{\rho_c(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}, \tag{3}$$

where ρ_c is the density of the core orbital hole. We can expand each molecular orbtial in the AO basis,

$$\phi_p(\mathbf{r}) = \sum_{\alpha} C_{\alpha p} \chi_\alpha(\mathbf{r}), \tag{4}$$

giving the AO representation of the matrix element

$$\langle \phi_p | \hat{v}_{\mathrm{ch}} | \phi_q \rangle, \approx \sum_{\alpha,\beta} C_{\alpha p}^* V_{\alpha\beta} C_{\beta q}, \tag{5}$$

where $V_{\alpha\beta}$ is the AO representation of the core-hole potential v_{ch},

$$V_{\alpha\beta} = -\iint d\mathbf{r}\, d\mathbf{r}' \frac{\chi_\alpha^*(\mathbf{r}) \chi_\beta(\mathbf{r}) \rho_c(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}. \tag{6}$$

Expanding the core-orbital ϕ_c (in the density ρ_c) in an AO basis contribution,

$$\phi_c(\mathbf{r}) = \sum_{\gamma} C_{\gamma c} \chi_\gamma(\mathbf{r}), \tag{7}$$

we arrive at an expression for the core-hole potential as a regular two-electron integral,

$$V_{\alpha\beta} = -\sum_{\gamma\delta} C_{\gamma c}^* C_{\delta c} (\alpha\beta|\gamma\delta), \tag{8}$$

where,

$$(\alpha\beta|\gamma\delta) = \iint d\mathbf{r}\, d\mathbf{r}' \frac{\chi_\alpha^*(\mathbf{r}) \chi_\beta(\mathbf{r}) \chi_\gamma^*(\mathbf{r}') \chi_\delta(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}. \tag{9}$$

Thus giving the total matrix element in the AO representation,

$$\boxed{\langle \phi_p | \hat{v}_{\mathrm{ch}} | \phi_q \rangle \approx -\sum_{\alpha,\beta} \sum_{\gamma,\delta} C_{\alpha p}^* C_{\beta q} C_{\gamma c}^* C_{\delta c} (\alpha\beta|\gamma\delta)} \tag{10}$$

In MO representation this is simply the four-center MO integral

$$\boxed{\langle \phi_p | \hat{v}_{\mathrm{ch}} | \phi_q \rangle \approx (pq|cc)} \tag{11}$$